

A fine balance: Exploring the interplay of reproductive strategies and growth dynamics in trees

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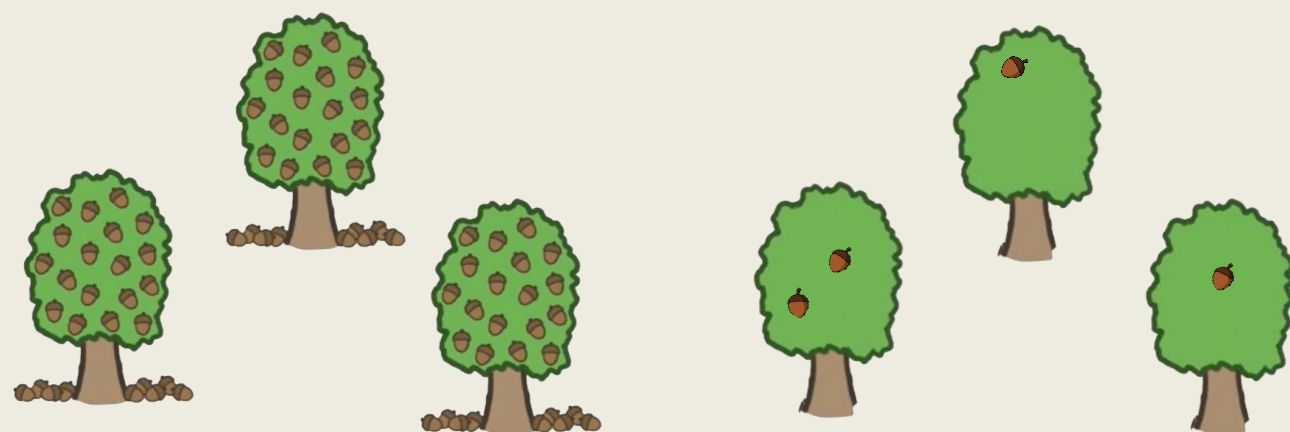
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Introduction

Masting—the phenomenon of synchronized, highly variable (“boom” and “bust” years) seed production across plant populations is believed to shape plant regeneration and seed-predator dynamics, yet still remains poorly predicted.

Mast year

Non-mast year



One hypothesized driver is a trade-off between vegetative growth and reproduction. However, it is often masked by environmental variation acting on both.

Here, we leverage rarely available data for both seed reproduction and growth over time and space to test for evidence of this trade-off while accounting for variation in the environment. This poster presents a model-in-progress for my PhD dissertation, and I am looking for help to improve this model.

Main questions

- Is there any evidence of a growth-reproduction trade-off?

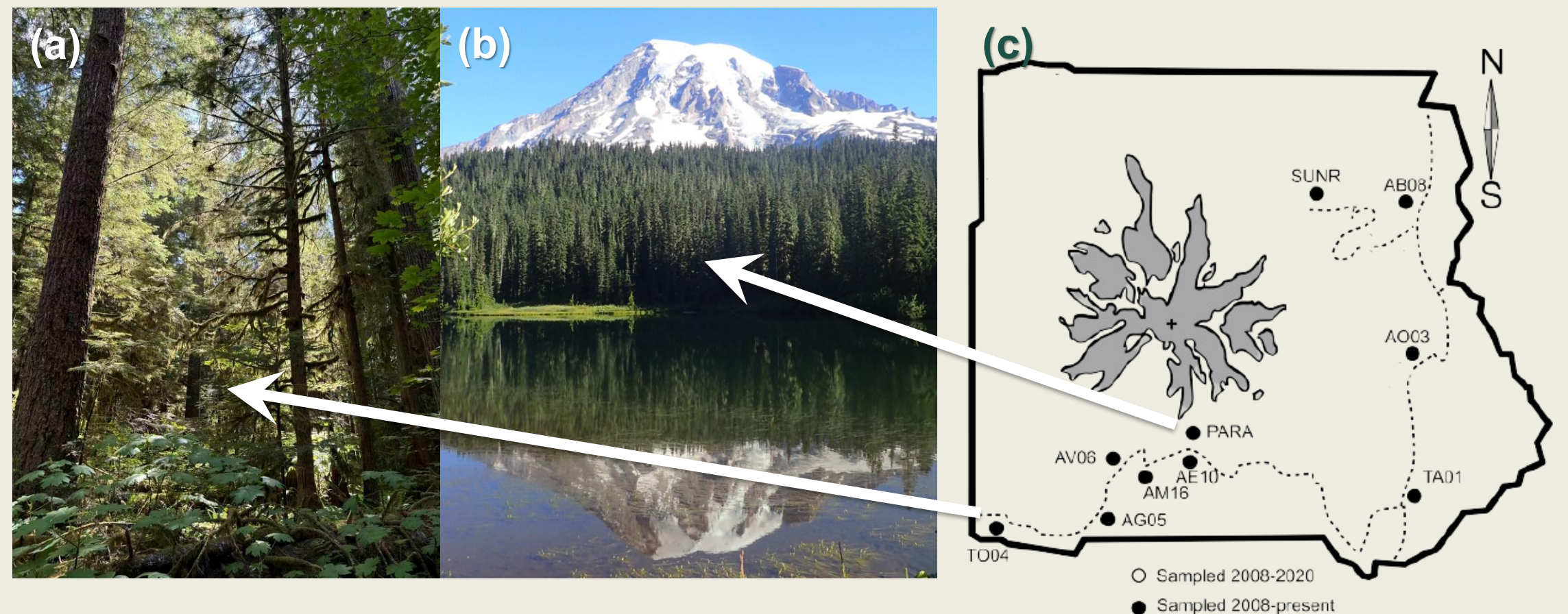
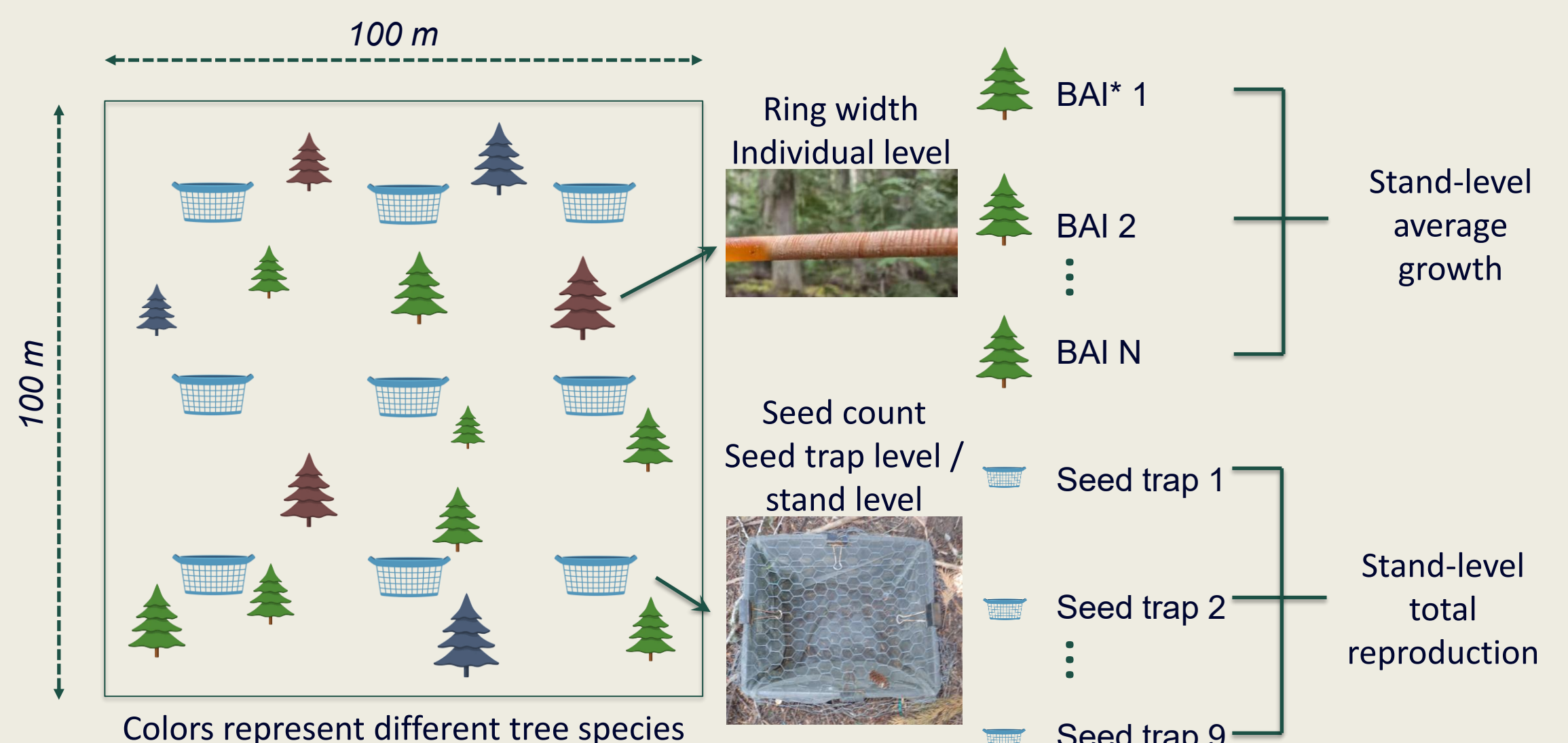


Figure 1. (a) Stand TO04 (elevation: ~700m) and (b) Paradise (PARA, elevation: ~1600m) at Mount Rainier (Tahoma) National Park, Washington, USA. (c) Map showing all the stands at different elevations. Each stand is a set of trees and seed traps and standardized to be 100m and 100m (see below).



*BAI: Basal area increment: $\left[\left(\frac{DBH_{y+1} - DBH_y}{2} + ring\ width \right)^2 - \left(\frac{DBH_y}{2} \right)^2 \right] * \pi$

Model

Growth

$$BAI_{i,t} \sim \text{Normal}(\alpha_{BAI}, \sigma_{BAI})$$

Each tree's annual growth (BAI) varies around a shared baseline growth (α_{BAI}), with some tree-to-tree and year-to-year variation (σ_{BAI}).

$$G_{i,t} = BAI_{i,t} - \alpha_{BAI}$$

For each tree, we calculate how much that year's growth ($G_{i,t}$) deviated from the baseline (α_{BAI}).

$$G_t = \text{mean}_i(G_i, t)$$

Averaged deviations across all trees in a stand to get one stand-level growth per year (G_t).

Reproduction

$$\text{Log}(\mu_{sc,t}) = \alpha_{sc} + \gamma_{\text{current}} \cdot G_t + \gamma_{\text{lag}} \cdot G_{t-1}$$

Expected seed production depends on a baseline level (α_{sc}), plus how this year's (γ_{current}) and last year's growth (γ_{lag}) affect the overall seed reproduction, accounting for an immediate trade-off and a lag effect.

$$sc_t \sim \text{LogNormal}(\text{Log}(\mu_{sc,t}), \sigma_{sc})$$

Observed seed counts vary around that expected value ($\text{Log}(\mu_{sc,t})$), reflecting natural variability (σ_{sc}) in reproduction beyond what growth can explain.

Preliminary results

One species (*Tsuga heterophylla*) at one stand (21 trees), from 2013 to 2024 (11 years):

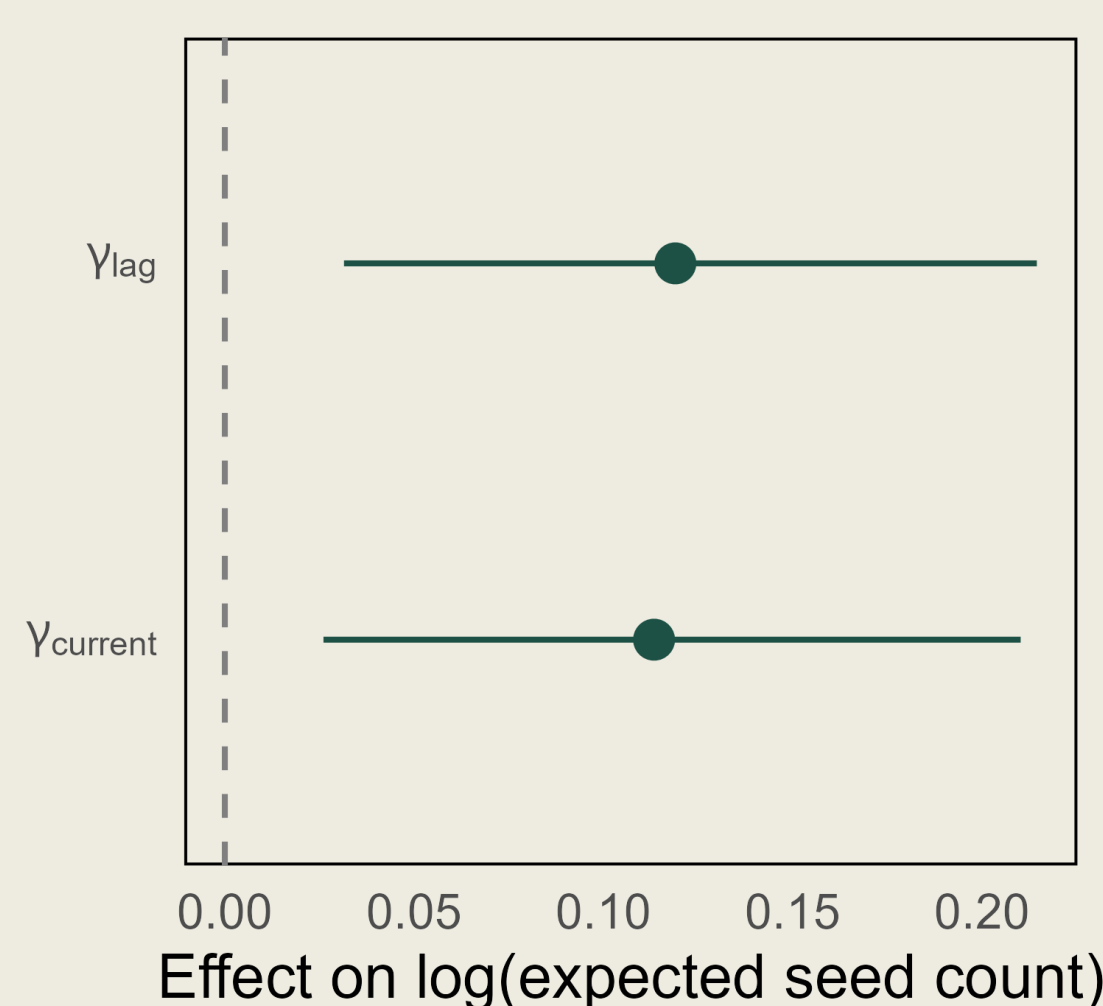


Figure 2. Posterior estimates of current-year (γ_{current}) and lagged (γ_{lag}) growth effects on seed production, with 95% credible intervals. Dashed line marks zero.

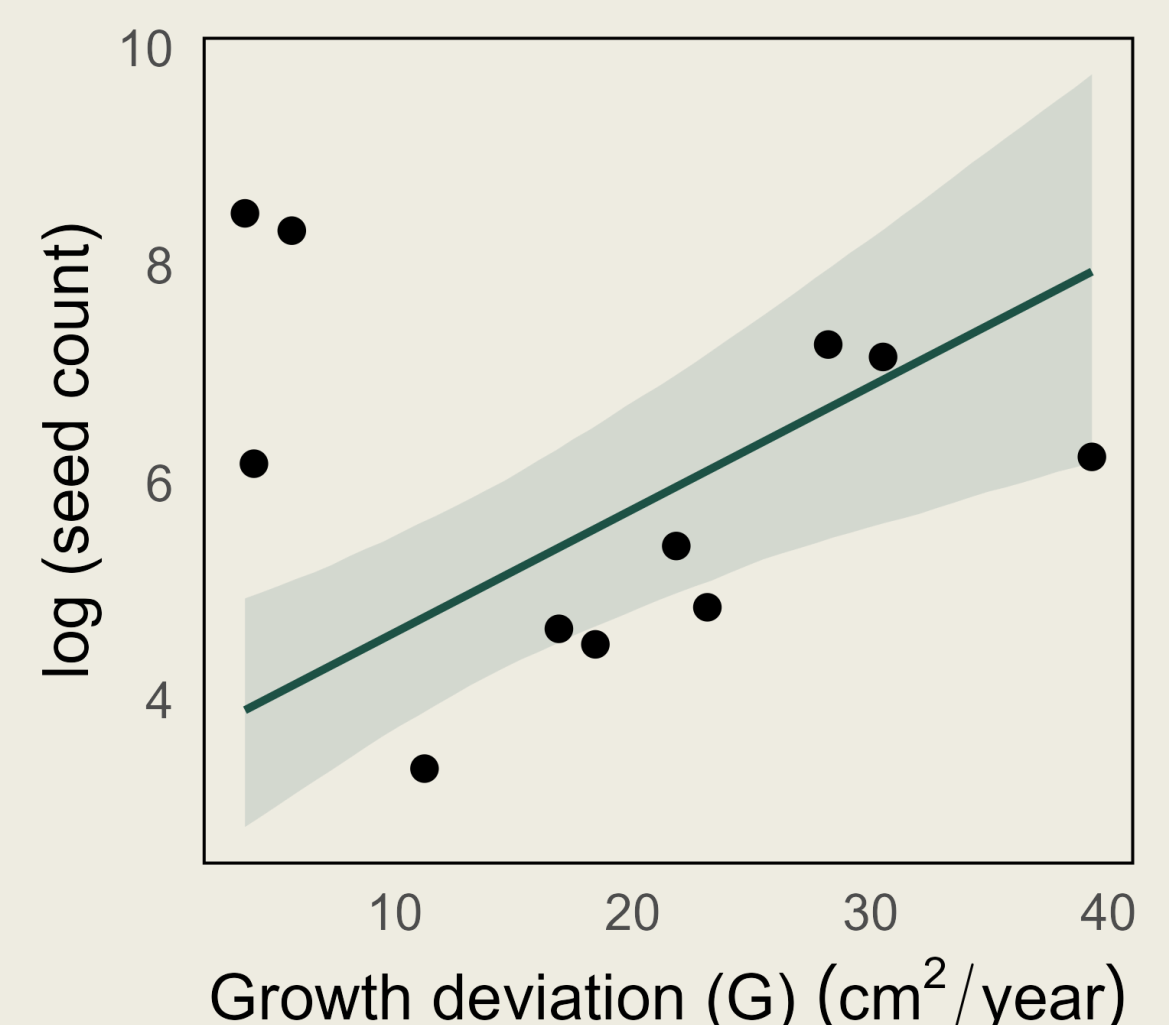


Figure 3. Predicted seed count (log scaled) as a function of growth deviation from baseline (green line = posterior mean, shaded band = 10–90% interval), with observed stand-level seed counts (black points). Prediction takes lagged growth at its average value.

Acknowledgements

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